# Prompt Compression and Token Budgeting for Enterprise Telemetry: Empirical Evaluation of LLMLingua-2 Control, Quality Tradeoffs, and Latency Bottlenecks

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**Abstract—** Deploying large language models for observability, incident triage, and postmortem synthesis requires processing dense diagnostic traces and technical documentation. This work evaluates LLMLingua-2 prompt compression and token budgeting using a calibrated controller and a curated synthetic benchmark of 60 incident cases. The committed artifact contains 240 runs across target ratios of 1x, 2x, 3x, and 5x using local Ollama-served Qwen 2.5 1.5B under deterministic decoding. Aggregate compression-ratio means approach requested targets, but per-case deviations remain. Quality is measured using task-specific exact match, token F1, ROUGE, key-fact recall, and semantic similarity (with historical backend provenance caveats). Compression adds several seconds of overhead in this local setup: end-to-end latency is approximately 9.9x baseline for Extractive QA at 5x and 2.97x across all tasks. The results demonstrate a measurable quality/latency tradeoff rather than a universal safe compression threshold.

**Index Terms—** Prompt Compression, LLMLingua-2, Token Budgeting, Large Language Models, Cloud Observability, Systems Efficiency.

## I. Introduction

Large language models are increasingly used in site reliability engineering and cloud infrastructure management. Diagnostic workflows combine Kubernetes events, system logs, distributed traces, and incident reports. Long contexts can increase prompt-processing cost and latency, while important evidence may be diluted by irrelevant material. Selective prompt compression offers an alternative to simple truncation by estimating token utility and removing lower-value content.

This study focuses on three questions: (1) how reliably can a requested compression ratio be controlled on structured incident contexts; (2) how does compression affect extractive QA, multi-hop reasoning, and summarization; and (3) how does compressor overhead compare with downstream inference savings in a local deployment?

The middleware preserves system instructions and user questions as invariants, applies chunking and question-aware ranking to context, compresses the selected context, enforces a token budget, and records telemetry. The contribution is an empirical characterization of the controller and its tradeoffs, not a claim that one ratio is universally safe.

## II. Middleware Design

**Context isolation.** System instructions and the user question are anchored and preserved. Compression is applied to document/log context only. This avoids modifying task instructions or query wording.

**Context selection.** Input is divided into token windows (800-token chunks with 80-token overlap) and ranked using lexical overlap with the question. Selected context is passed to LLMLingua-2.

**Ratio control.** The implementation uses an empirical power-law initial retention mapping, r■(R) = R■■·■■, followed by a bounded proportional refinement when the initial measured ratio deviates beyond the configured tolerance. This is a calibration heuristic; it does not guarantee an exact per-request ratio.

**Budget enforcement and telemetry.** The middleware applies a token-budget guard and records compression, inference, end-to-end latency, token counts, and configured cost estimates.

## III. Dataset and Experimental Setup

The benchmark consists of 60 curated synthetic incident cases, evenly divided among Extractive QA, Multi-Hop QA, and Incident Summarization. The scenarios are designed to resemble operational telemetry and postmortem material; they are not production incident records. Each case was evaluated once at each of four target ratios, yielding 240 rows. Thus, standard deviations describe variation across cases, not repeated-run uncertainty.

| Task | Cases | Primary metric | Additional metrics |
|---|---|---|---|
| Extractive QA | 20 | Exact Match | Token F1 |
| Multi-Hop QA | 20 | Token F1 | ROUGE-L, semantic similarity, key-fact recall |
| Summarization | 20 | ROUGE-L | ROUGE-1/2, key-fact recall |

**Model and runtime.** All 240 CSV rows identify qwen2.5:1.5b. Generation used Ollama /api/generate with temperature 0.0, top-p 1.0, and streaming disabled. Model metadata reports Qwen2 architecture, Q4_K_M quantization, approximately 1.54B parameters, a

32,768-token native context, and 1,536-dimensional embeddings. The benchmark context limit was 4,096 tokens.

**Machine and software.** MacBook Air (Mac14,2), Apple M2, 8 CPU cores, 8 GB RAM, ARM64, macOS 26.6.2. Reported versions: Python 3.13.7, Ollama 0.40.11, PyTorch 2.6.0, Transformers 5.20.2, tiktoken 0.14.0, and LLMLingua 0.2.2. PyTorch MPS was available; however, the CSV does not record whether compression executed on CPU or MPS, so no device attribution is made.

## IV. Compression-Ratio Results

| Target | Mean actual | SD | Within ±5% | Within ±15% |
|---|---|---|---|---|
| 1.0x | 1.000 | 0.000 | 60/60 | 60/60 |
| 2.0x | 1.980 | 0.152 | 25/60 | 58/60 |
| 3.0x | 2.984 | 0.215 | 25/60 | 60/60 |
| 5.0x | 4.837 | 0.352 | 27/60 | 60/60 |

The calibrated controller produces aggregate means close to requested ratios. Nevertheless, only 25 of 60 cases at 2x, 25 at 3x, and 27 at 5x fall within ±5% of the target. Aggregate alignment must not be interpreted as a per-request guarantee.

## V. Task Quality

| Task / metric | 1x | 2x | 3x | 5x |
|---|---|---|---|---|
| Extractive EM | 0.750 | 0.550 | 0.500 | 0.350 |
| Extractive F1 | 0.890 | 0.650 | 0.533 | 0.383 |
| Multi-Hop F1 | 0.464 | 0.400 | 0.352 | 0.353 |
| Multi-Hop ROUGE-L | 0.391 | 0.328 | 0.285 | 0.280 |
| Multi-Hop semantic similarity* | 0.825 | 0.803 | 0.783 | 0.736 |
| Multi-Hop key-fact recall | 0.531 | 0.413 | 0.360 | 0.334 |
| Summarization ROUGE-L | 0.319 | 0.267 | 0.261 | 0.232 |
| Summarization key-fact recall | 0.481 | 0.353 | 0.307 | 0.250 |

*The original CSV does not record semantic-similarity backend provenance. The implementation supports MiniLM embeddings and a token-Jaccard fallback; historical values cannot be attributed exclusively to MiniLM.

Extractive QA quality declines steadily with stronger compression. Multi-Hop QA metrics also decline overall, with a small near-flat change in F1 between 3x and 5x. Summarization ROUGE-L and key-fact recall decline as compression increases. These are descriptive results over the selected synthetic cases.

## VI. Latency and Cost

| Task | Ratio | Compression ms | Inference ms | E2E ms |
|---|---|---|---|---|
| Extractive QA | 1x | 0.4 | 411.7 | 412.1 |
| Extractive QA | 5x | 3,839.9 | 254.9 | 4,094.8 |
| Multi-Hop QA | 1x | 0.4 | 1,970.7 | 1,971.1 |
| Multi-Hop QA | 5x | 4,920.3 | 1,309.4 | 6,229.7 |
| Summarization | 1x | 0.4 | 3,406.1 | 3,406.5 |
| Summarization | 5x | 5,466.9 | 1,410.7 | 6,877.6 |

Across all 60 cases, mean end-to-end latency increased from 1,929.9 ms at 1x to 5,734.0 ms at 5x (2.97x). Mean inference latency decreased from 1,929.5 ms to 991.7 ms, but compression overhead outweighed the inference savings. The CSV does not record compressor execution device, so the observed overhead is not attributed to CPU or MPS.

All recorded estimated cost fields are zero under the local Ollama configuration. Therefore, this experiment does not measure commercial API dollar savings.

## VII. Limitations and Interpretation

The dataset is curated and synthetic, and each case/ratio has only one run. No repeated-run confidence intervals or significance tests are available. The original CSV omits semantic metric backend provenance and compressor execution device. The measured hardware, software, model, and prompt workload are specific; results should not be generalized to other systems without replication. No universal safe ratio or commercial cost benefit is established.

Future work should add externally sourced incident datasets, repeated trials, explicit hardware-device telemetry, persisted semantic-metric backend identifiers, and evaluation across additional model sizes and runtimes. Human review of retained facts and failure cases would further strengthen deployment guidance.

## VIII. Conclusion

This evaluation shows that calibrated prompt compression can bring aggregate compression ratios close to their requested values, while meaningful per-case deviations remain. Stronger compression reduces context size and downstream inference time but degrades several task-quality metrics and increases end-to-end latency in the measured local environment. The appropriate compression setting is therefore workload- and risk-dependent and must be validated on representative data before deployment.

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